

LightGBM-SHAP Ensemble for Early Dropout Detection in Ghanaian Basic Schools:

A CTGAN-Augmented Institutional Data Framework with LightGBM, SHAP TreeExplainer, and Teacher-Validated Explanations

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Field	Entry
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★ NOVELTY DECLARATION

Crisp Novelty Statement:

This is the first study to apply a LightGBM-SHAP ensemble to basic school dropout prediction in Ghana, using a novel CTGAN-augmented institutional dataset validated against real Ghanaian school distributions. It further evaluates whether SHAP TreeExplainer outputs are actionable for teachers and district officers operating under low-resource, no-LMS conditions, and an evaluation dimension absent from all prior sub-Saharan African EdTech ML work.

Falsifiability: If a study can be identified applying LightGBM or an equivalent gradient-boosted model with SHAP explainability to Ghanaian or sub-Saharan African basic school administrative records, this novelty claim would be narrowed to the fairness evaluation and teacher-validation dimensions.

Novelty Types:

- ✓ New Application Context: first LightGBM-SHAP application to basic school dropout in Ghana/SSA
- ✓ New Dataset Contribution: original institutional dataset from Ghanaian Primary and JHS schools; not OULAD-dependent, with CTGAN augmentation built from real data distributions

- ✓ New Evidence in Under-Researched Population / Geography: sub-Saharan African basic school pupils, ages 6–15, no-LMS environment
- ✓ HCAI User Study: teacher and GES district officer validation of SHAP explanation actionability; absent from all prior SSA EdTech ML work

Note on dataset independence: OULAD (enriched) serves only as a structural plausibility reference for CTGAN calibration. This study contributes an original dataset; it is not an OULAD-dependent study.

SECTION 1: PROBLEM STATEMENT

1.1 Background

Dropout from basic education, spanning Primary 1 to Junior High School Form 3 (ages 6–15), remains a critical development challenge in Ghana. Under the Free Compulsory Universal Basic Education (FCUBE) policy, basic schooling is constitutionally guaranteed. Yet the Ghana Education Service (GES) reports gross JHS dropout rates exceeding 20%, with disproportionate attrition in rural and northern districts (MoE, 2022). These causal drivers, which include poverty, child labor, food insecurity, travel distance, and early pregnancy among girls, are factors that interact in complex, nonlinear ways that manual monitoring systems cannot detect in time for preventive intervention (MoE, 2022).

Existing early warning mechanisms are reactive: headteachers identify at-risk pupils only after prolonged absence or academic failure, by which point re-enrolment becomes significantly harder. Limited intervention resources, such as school feeding funds, LEAP social protection payments, and bursaries, are allocated without evidence-based pupil-level targeting. This represents a critical inefficiency with high human cost.

1.2 Specific Problem and Gap

Gradient-boosted tree models have emerged as the state-of-the-art for tabular classification in educational data mining, with LightGBM (Ke et al., 2017) offering particular advantages over XGBoost and Random Forest for medium-scale, heterogeneous datasets: faster training via histogram-based leaf-wise splitting, native handling of categorical features, and strong performance under class imbalance all characteristics directly matching the Ghanaian basic school administrative dataset. A March 2025 Scientific Reports study (Marcolino et al., 2025) confirmed that CatBoost+SHAP ensembles achieve Tier 1 publication acceptance for dropout prediction; LightGBM occupies the same model class with comparable or superior performance on tabular data of this type. Critically, LightGBM is natively compatible with SHAP TreeExplainer, enabling per-pupil attribution scores with no additional architectural complexity.

SHAP (SHapley Additive exPlanations) has emerged as the leading XAI technique for rendering ML predictions interpretable to non-technical stakeholders (Lundberg & Lee, 2017). However, the SHAP application in African basic education is absent from the literature. No study has applied a GNN with SHAP explainability to Ghanaian or sub-Saharan African basic school dropouts using institutional administrative data. Namoun & Alshantiti (2021) note that explainability is absent from 88% of dropout papers; none of the studies they reviewed evaluated stakeholder interpretability in a low-income, no-LMS school context.

This gap is the precise contribution this study addresses: a LightGBM-SHAP ensemble trained on Ghanaian basic school administrative data, with SHAP TreeExplainer outputs validated for actionability by teachers and GES district officers.

1.3 Research Objectives and Research Questions

Objective 1a: To develop a LightGBM primary model paired with a Random Forest secondary model for predicting dropout risk among Primary and JHS pupils in Ghanaian basic schools using a CTGAN-augmented institutional dataset.

Objective 1b: To evaluate the LightGBM-SHAP ensemble against Logistic Regression, Decision Tree, and XGBoost baselines on AUC-ROC, AUC-PR, and macro F1-score.

RQ1: How accurately does the LightGBM-SHAP ensemble predict dropout risk among Ghanaian basic school pupils compared to Logistic Regression, Decision Tree, and XGBoost baselines, as measured by AUC-ROC, AUC-PR, and macro F1-score?

Objective 2: To identify, via SHAP TreeExplainer analysis, which student, household, and school-level features most significantly predict dropout risk and are actionable for Ghanaian teachers and GES district officers.

RQ2: Which features identified by SHAP analysis are rated as most actionable by Ghanaian basic schoolteachers and GES district officers for targeting dropout-prevention interventions?

Objective 3: To evaluate whether the dropout prediction model exhibits equitable performance across gender and geographic zone (urban vs. rural) subgroups using fairness metrics.

RQ3: How do Equal Opportunity and Equalized Odds fairness gaps compare across gender and urban/rural subgroups in the trained LightGBM dropout prediction model?

1.4 Theoretical Framework

This study is grounded in Bronfenbrenner's Ecological Systems Theory (1979), which holds that children's educational outcomes are shaped by nested environmental systems: student, household, school, and community. This directly motivates the multi-source feature design: a model using only academic performance scores is theoretically insufficient; household poverty proxies, school feeding participation, district deprivation indices, and geographic distance must all be incorporated as input features. LightGBM's ability to handle heterogeneous tabular features drawn from all four ecological system levels without requiring manual normalisation of interactions between levels directly operationalises this theoretical requirement.

Self-Determination Theory (Deci & Ryan, 1985) informs the HCAI dimension: SHAP-explained predictions communicated to teachers support educator 'competence' by enabling targeted, evidence-based intervention rather than stigmatizing blanket labelling, positioning the system as a tool that augments human educational agency. Both Bronfenbrenner and SDT are explicitly referenced again in the Discussion section to demonstrate theoretical contribution, not only theoretical motivation.

SECTION 2: RELATED WORK

Existing ML dropout research clusters around four limitations relevant to this study.

Models and Approaches

Existing ML approaches to dropout prediction have converged on gradient-boosted tree ensembles as the state-of-the-art for tabular educational data (Mduma et al., 2023; Fernández-García et al., 2022; Chango et al., 2021). SHAP has emerged as the leading post-hoc explainability method for these models (Lundberg & Lee, 2017; Namoun & Alshanqiti, 2021). Yet this methodological convergence has occurred entirely within OECD university and fMOOC contexts; the structural features driving dropout in sub-Saharan African basic schools, such as poverty proxies, seasonal attendance loss, and grade repetition, are absent from every benchmark dataset used to validate these methods. This methodological convergence has occurred entirely within OECD university and MOOC contexts. The structural features driving dropout in sub-Saharan African basic schools, such as poverty proxies, seasonal attendance loss, grade repetition, are absent from every benchmark dataset used to validate these methods. A 2025 Tier 1 study (Marcolino et al., 2025) demonstrated that CatBoost+SHAP achieves strong AUC-ROC on dropout prediction with teacher-interpretable explanations; LightGBM occupies the same model class and offers additional speed and categorical feature handling advantages. No prior study applies LightGBM-SHAP to sub-Saharan African basic school dropout, creating both a model-class gap and a geographic gap that this study addresses simultaneously.

Explainability Gap

Namoun & Alshanqiti (2021) found that fewer than 12% of dropout prediction papers include any explainability mechanism. Where feature importance is reported, it is the global aggregate from tree-based models, not the per-prediction SHAP explanations needed for individual teacher decisions. Lundberg & Lee (2017) establish

SHAP as theoretically superior to simpler importance rankings. No study has applied SHAP to dropout prediction in an African basic school context or evaluated whether SHAP outputs are interpretable by Ghanaian educators with limited ML literacy.

African and Basic School Gap

Adekitan & Salau (2019) applied Random Forest to Nigerian university records, the closest African ML precedent, without XAI and without basic school data. Psaki et al. (2022) document dropout patterns in sub-Saharan African basic education using survey statistics, not ML models. Adewoyin & Igboeli (2022) identify the absence of equity-aware, explainable AI in African education as a critical research gap. This study fills that gap at the basic school level in Ghana, with a LightGBM-SHAP ensemble, teacher-validated explanations, and explicit fairness evaluation.

Fairness in EdTech ML

A parallel literature on algorithmic fairness in education (Kizilcec & Lee, 2022; Baker & Hawn, 2022) warns that predictive models trained on historically under-resourced populations can amplify existing inequities. No prior sub-Saharan African dropout study has evaluated Equal Opportunity or Equalized Odds fairness gaps. This makes the fairness evaluation component of this study a distinct contribution independent of the predictive accuracy results, addressing a recognized gap in both the African EdTech literature and the global algorithmic fairness literature.

Three comparison baselines are selected: Mduma et al. (2023): XGBoost/RF on sub-Saharan African university dropout; Chango et al. (2021): multi-source ensemble model on tabular student data; Fernández-García et al. (2022): systematic review identifying SHAP as the best explainability method.

All three represent state-of-the-art tabular approaches and are appropriate baselines for evaluating the LightGBM-SHAP ensemble on the same Ghanaian dataset.

SECTION 3: METHODOLOGY

3.1 Primary Model: LightGBM + Random Forest Ensemble with SHAP TreeExplainer

This study adopts LightGBM (Ke et al., 2017) as the primary prediction model, paired with a Random Forest secondary model and SHAP TreeExplainer as the XAI layer. This architecture is confirmed as Tier 1-viable by Marcolino et al. (2025) who achieved publication in Scientific Reports using CatBoost+SHAP for dropout prediction. LightGBM occupies the same model class with equivalent or superior performance on medium-scale tabular data.

- (i) pupil-to-classroom membership
- (ii) classroom-to-school membership
- (iii) LightGBM is selected as the primary model for three reasons specific to this dataset:
 - (i) Histogram-based leaf-wise splitting achieves faster training and equivalent or superior AUC-ROC on medium-scale tabular datasets.
 - (ii) native handling of categorical features without one-hot encoding overhead.
 - (iii) built-in class imbalance handling via `is_unbalance=True`.
- (i) The Random Forest secondary model provides an independent probability estimate that, when averaged with LightGBM output, reduces variance caused by individual model sensitivity to hyperparameter choices, aligning with Breiman (2001) ensemble variance-reduction principles. This ensemble averaging is the main innovation over using LightGBM alone and is directly tested in ablation studies. SHAP TreeExplainer is chosen because it offers exact (not approximate) Shapley value calculations natively compatible with both LightGBM and Random Forest, enabling:
 - (i) global SHAP summary beeswarm plots for GES policy briefings; and
 - (ii) local waterfall plots for individual teacher decisions on flagged pupils. LIME is included as an alternative explainer for sensitivity comparison. The full sentence was truncated in the

original; SHAP TreeExplainer is preferred because it provides precise Shapley value computation (Lundberg & Lee, 2017), enabling:

- (i) global feature importance summary plots for GES policy briefings; and
- (ii) local attribution plots for individual predictions to inform teacher decisions. LIME remains as an alternative for sensitivity analysis across both models.

3.2 Dataset: Institutional Data + CTGAN Synthetic Augmentation

This study uses a two-component dataset strategy, as neither OULAD nor EdNet contains the specific feature set of Ghanaian basic schools.

- **Primary source:** Institutional administrative records collected from ≥ 10 basic schools across 2 districts in the Ashanti Region of Ghana, accessed via GES EMIS and individual school registers. School selection will follow purposive sampling to maximize geographic variation within the Ashanti Region, with at least 5 urban Ashanti schools and 5 peri-urban Ashanti schools across the 2 selected districts. This design directly tests model generalisability across the urban/peri-urban dimension targeted in RQ3. Minimum real-record target: 1,000 records (≥ 350 dropout cases, ≥ 650 retained). With 10 schools averaging 100–200 registered JHS pupils per school, this target is operationally achievable. The GES EMIS access request will be submitted simultaneously with the CHRPE application. Paper registers will be digitised by the data collection team using standardised entry forms; estimated digitisation capacity is 50 records per person per day (4-person team: 200 records/day).
- **Multi-year records:** Where GES EMIS records span multiple academic years per pupil, data will be structured at the pupil-year level (one row per pupil per year). A pupil present in 3 consecutive school years yields 3 data rows, effectively tripling the real dataset volume while maintaining data authenticity. School-level contextual features (pupil-teacher ratio, infrastructure score, school feeding days per term) will be added from GES EMIS as additional input features, theoretically motivated by Bronfenbrenner's mesosystem layer.
- **Synthetic augmentation:** If collected records fall below 1,500 after preprocessing, CTGAN (via SDV 1.8) will generate synthetic records from the real data distribution. All synthetic records will be explicitly labelled as synthetic. Validation: Level 1 (SDMetrics statistical fidelity); Level 2 (TSTR vs TRTR comparison, essential); Level 3 (NNAA privacy audit); Level 4 (GES teacher plausibility review of 20 synthetic records). SMOTE is applied to the training set only for class imbalance handling.

3.3 Preprocessing and Feature Engineering

Numerical features (attendance rate, exam scores) will be standardized using StandardScaler. Categorical features (gender, school type, district zone) will be one-hot encoded. Missing values will be imputed using median strategy for numerical and mode for categorical features. Dropout is defined per the GES operational definition: absence from school for 3 or more consecutive terms without transfer. The binary outcome label (dropout = 1, retained = 0) will be derived from school register records. The dataset is cross-sectional at the pupil-year level; each record represents one academic year per pupil. No temporal leakage risk applies as records are not ordered time series within a single academic year.

3.4 Experimental Design

Item	Plan
Data split	Stratified 70/15/15 train/validation/test; stratified by dropout label and district zone
Model architecture	Pupil nodes (admin features); classroom nodes (aggregate features); school nodes (infrastructure, feeding days, deprivation index). Edges: pupil→classroom membership; classroom→school membership; pupil→pupil same-grade proximity
Class imbalance	SMOTE on training set only; class_weight='balanced' in all tabular baseline models
Evaluation protocol	5-fold stratified cross-validation + held-out test set; mean $\pm$ SD reported across folds and across 5 random seeds (42, 123, 456, 789, 1024) — required for Tier 1 result stability reporting
Hyperparameter tuning	LightGBM primary: num_leaves $\in \{31, 63, 127\}$, learning_rate $\in \{0.01, 0.05, 0.1\}$, n_estimators $\in \{200, 500, 1000\}$, is_unbalance=True. Random Forest secondary:

XAI implementation	n_estimators $\in \{100, 200, 300\}$, max_features $\in \{\text{sqrt}, \text{log2}\}$. Final prediction: weighted average of LightGBM and RF probability outputs. Baselines: LR, Decision Tree, XGBoost (n_estimators $\in \{100, 200, 300\}$, max_depth $\in \{4, 6, 8\}$)
Baselines	SHAP TreeExplainer (exact Shapley values; native LightGBM and RF compatibility); global beeswarm summary plot; local waterfall plots for 5 high-risk cases; validated for actionability by 2 GES teachers and 1 district officer by 2 GES teachers and 1 district officer
Statistical tests	Logistic Regression · Random Forest · XGBoost (Chen & Guestrin, 2016) · Baseline 1 (Mduma et al., 2023) · Baseline 2 (Chango et al., 2021) · Baseline 3 (Fernández-García et al., 2022)
Fairness metrics	McNemar's test and Wilcoxon signed-rank test; $\alpha = 0.05$; 95% CIs reported for all key comparisons
Ablation study	Equal Opportunity gap and Equalized Odds gap by gender and urban/rural zone; if gap > 5%, fairness-constrained retraining recommended
Checkpoint saving	(i) LightGBM alone vs LightGBM+RF ensemble — quantifies value of ensemble combination · (ii) No XAI layer (iii) No socioeconomic features (iv) Real data only vs CTGAN-augmented
Computing environment	Model checkpoints saved to Google Drive at end of each CV fold using joblib.dump(); Drive-mounted path /content/drive/MyDrive/ghana_dropout_ckpts/ used throughout to prevent Colab session loss
Random seeds	Google Colab Pro (standard GPU sufficient — LightGBM does not require A100); Kaggle Notebooks for reproducibility kernel
Software versions	5 seeds: 42, 123, 456, 789, 1024; mean $\pm$ SD reported across seeds for all metrics
Reproducibility artefacts	Python 3.10 · LightGBM 4.3 · scikit-learn 1.4 · SHAP 0.44 · SDV 1.8 (CTGAN) · pandas 2.1 · fairlearn 0.10
	requirements.txt (pinned versions) and conda environment.yml committed to repository root before data collection begins

SECTION 4: PLANNED RESULTS & EVALUATION METRICS

The following metrics will be reported for all models (LightGBM-RF ensemble, Logistic Regression, Decision Tree, XGBoost, and three literature baselines (Mduma et al., 2023; Chango et al., 2021; Fernández-García et al., 2022)). Truncated continuation: The following metrics will be reported for all model literature baselines):

- AUC-ROC: primary classification metric (appropriate for imbalanced dropout dataset)
- AUC-PR (Precision-Recall Curve): more informative than ROC when the positive class (dropout) is rare.
- Macro F1-Score: balances precision and recall across imbalanced classes.
- Precision and Recall: reported separately; recall prioritized (minimizing missed at-risk pupils)
- SHAP TreeExplainer global summary (beeswarm plot) and local waterfall plots for 5 selected high-risk cases
- Fairness: Equal Opportunity gap and Equalized Odds gap by gender and urban/rural zone
- Statistical significance: McNemar's test, Wilcoxon signed-rank test, p-values and 95% CIs for all pairwise model comparisons.

Planned visualization outputs: ROC curves for all models on one figure; Precision-Recall curves; SHAP beeswarm summary; SHAP waterfall plots (5 cases); SHAP dependence interaction plots for top feature pairs (e.g. attendance rate $\times$ poverty proxy); normalised confusion matrices; learning curves; ablation study table; fairness comparison table; statistical significance table; synthetic data validation report (SDMetrics + TSTR comparison). Full text was truncated here: Planned visualization outputs: ROC curves for all models on one figure; Precision-Recall curves; SHAP beeswarm summary; SHAP attribution plots (5 cases); SHAP

dependence plots for top 3 features; normalized confusion matrices; learning curves (train vs validation loss over epochs); ablation study table; fairness comparison table; statistical significance table; synthetic data validation report (SDMetrics + TSTR comparison); graph visualization of school-cohort network with predicted risk scores overlaid on node color.

SECTION 5: ETHICS, DISCUSSION PLAN & TIMELINE

5.1 Ethical Clearance Plan

This study collects new administrative data from pupils and school records. KNUST CHRPE clearance is required before any data collection. Application will be submitted under the Social Science category (contact: chrpe@knust.edu.gh; fee approximately GHS 100–200; turnaround approximately 2–6 weeks). Written informed consent will be obtained from parents/guardians; assent from pupils aged 10+; headteacher and GES district director formal authorization. All pupil identifiers will be replaced with study codes. Data stored on an encrypted KNUST institutional Google Drive, restricted to the research team. All results presented at aggregate and anonymized individual levels. Full compliance with the Ghana Data Protection Act (2012).

Timeline buffer: Phase 3 (Data Collection, Weeks 4–8) is scheduled to begin only after CHRPE approval. One additional buffer week has been built between expected CHRPE approval and planned data collection start, to account for the possibility of the full 6-week approval timeline. If CHRPE approval is delayed beyond Week 5, Phase 3 will shift accordingly, compressing the Phase 7 writing period rather than beginning data collection without clearance.

Ethics statement for paper: This study was approved by the KNUST Committee on Human Research, Publications and Ethics (CHRPE), Reference No: [to insert upon receipt]. All data were collected with informed consent and anonymized in compliance with the Ghana Data Protection Act (2012).'

5.2 Discussion Plan

The discussion will address all required elements:

- (i) interpretation of whether the LightGBM-RF ensemble outperforms individual baselines, and whether the advantage is attributable to the ensemble combination, the socioeconomic feature set, or the CTGAN augmentation, or ablation design directly tests each component. Full text was truncated here: interpretation of whether the LightGBM-RF ensemble outperforms individual baseline graph structure specifically (ablation design directly tests this).
- (ii) alignment with Mduma et al. (2023) and Chango et al. (2021) benchmarks with structural explanations for divergence.
- (iii) honest limitations: single-region data, cross-sectional pupil-year design, recording bias in teacher-reported attendance, and potential generalizability constraints to northern Ghana.
- (iv) practical contribution: SHAP outputs reformatted into a 1-page teacher briefing template in English and Twi, showing the top 3 risk factors per flagged pupil, piloted with 5 teachers across 2 partner schools.
- (v) policy contribution: findings inform GES pupil targeting recommendations for the Ghana School Feeding Programme and LEAP payment allocation.
- (vi) SDG alignment: SDG 4 (Quality Education), SDG 10 (Reduced Inequalities), SDG 3 (Good Health).
- (vii) fairness analysis: if the Equal Opportunity gap exceeds 5% across any subgroup, fairness-constrained recommendations will be proposed.
- (viii) explicit theoretical contribution: both Bronfenbrenner and SDT are revisited to confirm that the multi-source feature design and SHAP outputs contributed to theory, specifically that Bronfenbrenner's multi-level ecological features demonstrably improve prediction over single-level academic features alone (tested via the no-socioeconomic-features ablation). Truncated continuation: explicit theoretical contribution: both Bronfenbrenner and SDT are revisited to confirm that the multi-source feature design and SHAP outputs contributed to theory, not only to prediction accuracy.

5.3 Journal Selection and APC

Journal	Details
Target Journal 1	Computers & Education: AI (Elsevier): APC approximately USD 2,400. Ghana qualifies for a Research4Life waiver. Apply at researchforaccess.com before submission. Scope: AI/ML in education, HCAI, and early warning systems. Review turnaround: 2–3 months. Download a 2023–2025 sample paper before submission to align formatting.
Target Journal 2 (backup)	Education and Information Technologies (Springer): APC approximately USD 3,190. Springer LMIC waiver available at springernature.com/gp/openresearch. Scope: broad EdTech, Ghana-context studies welcomed. Review turnaround: 2–4 months.

5.4 Timeline

Phase	Activity	Duration	Start	End
1: Prep	Ethics clearance (with 1-week buffer before data start); GitHub setup; requirements.txt + environment.yml committed; GES consent letters	3 weeks + 1 buffer	Wk 1	Wk 4
2: Literature	Complete 10–15 paper table; synthesis; verify DOIs; confirm 3 baselines	2 weeks	Wk 2	Wk 4
3: Data	Collect school records via GES EMIS + paper registers; digitise (est. 200 records/day with 4-person team); clean; CTGAN augmentation; SMOTE	4 weeks	Wk 5	Wk 9
4: Baselines	Implement tabular baselines (LR, RF, XGBoost); establish benchmark AUC-ROC	2 weeks	Wk 9	Wk 11
5: Model	Train LightGBM + RF ensemble; hyperparameter grid search; SHAP TreeExplainer implementation; ablation	3 weeks	Wk 11	Wk 14
6: Evaluation	Full metrics; fairness analysis; SHAP teacher validation; significance tests; multi-seed runs	2 weeks	Wk 14	Wk 16
7: Writing	Full paper draft in journal style; figures formatted	4 weeks	Wk 16	Wk 20
8: Review	Supervisor review; revisions; formatting; plagiarism check	2 weeks	Wk 20	Wk 22
9: Submit	Submit to Computers & Education: AI (Journal 1)	1 week	Wk 22	Wk 23

SECTION 6: CORE REFERENCES (APA 7, DOI-verified)

- Adekitan, A. I., & Salau, O. (2019). The impact of engineering students' performance on graduation using educational data mining. *Heliyon*, 5(2), e01250. <https://doi.org/10.1016/j.heliyon.2019.e01250>
- Adewoyin, Y., & Igboeli, G. (2022). AI for equity in African education: Challenges and opportunities. *AI & Society*. <https://doi.org/10.1007/s00146-022-01416-1>
- Baker, R. S., & Hawn, A. (2022). Algorithmic bias in education. *Journal of Learning Analytics*, 9(2), 1–23. <https://doi.org/10.18608/jla.2022.7615>
- Chango, W., Lara, J. A., Cerezo, R., & Romero, C. (2021). Improving prediction of students at risk of academic failure using multi-source data and MLP neural networks. *Future Generation Computer Systems*, 115, 479–491. <https://doi.org/10.1016/j.future.2021.07.013>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. <https://doi.org/10.1145/2939672.2939785>
- Fernández-García, A. J., et al. (2022). A systematic review of machine learning techniques for student dropout prediction. *IEEE Access*, 10, 33742–33762. <https://doi.org/10.1109/ACCESS.2022.3161757>

7. Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30. <https://doi.org/10.48550/arXiv.1711.08766> *ral Information Processing Systems*, 30. <https://doi.org/10.48550/arXiv.1706.02216>

8. Kabathova, J., & Drlik, M. (2021). Towards predicting students' dropout using different ML techniques. *Applied Sciences*, 11(7), 3130. <https://doi.org/10.3390/app11073130>

9. Kizilcec, R. F., & Lee, H. (2022). Algorithmic fairness in education. In C. Lang et al. (Eds.), *The Handbook of Learning Analytics* (2nd ed., pp. 167–177). SOLAR. <https://doi.org/10.18608/hla22.018>

10. Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30. <https://doi.org/10.48550/arXiv.1705.07874>

11. Mduma, N., Kalegele, K., & Machuve, D. (2023). Machine learning approaches for student dropout prediction in sub-Saharan Africa. *Education and Information Technologies*, 28, 3549–3571. <https://doi.org/10.1007/s10639-023-11604-7>

12. Ministry of Education, Ghana. (2022). Education sector annual performance report 2022. Ghana Education Service. <https://www.moe.gov.gh/education-sector-annual-performance-report-2022/>

13. Namoun, A., & Alshantqiti, A. (2021). Predicting student performance using data mining and learning analytics. *Applied Sciences*, 11(1), 237. <https://doi.org/10.3390/app11010237>

14. Psaki, S., McCarthy, K., & Mensch, B. (2022). Measuring and understanding gender differences in dropout in LMICs. *International Journal of Educational Development*, 88, 102608. <https://doi.org/10.1016/j.ijedudev.2022.102608>

15. UNESCO. (2023). Global education monitoring report 2023: Technology in education. UNESCO Publishing. ISBN 978-92-3-100612-4.

16. Marcolino, A. S., de Mello, R. F., & Wives, L. K. (2025). Predicting student dropout using CatBoost and SHAP explainability: A benchmark study. *Scientific Reports*, 15, 9840. <https://doi.org/10.1038/s41598-025-09840-x> *national Conference on Learning Representations*. <https://doi.org/10.48550/arXiv.1710.10903>

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Signature	
Date of Review	
Status	☐ Approved as submitted ☐ Approved with minor revisions ☐ Requires major revision ☐ Rejected — resubmit
Comments	